Code No. 3306

OSMANIA UNIVERSITY
FACULTY OF ENGINEERING
UNIVERSITY COLLEGE OF ENGINEERING (AUTONOMOUS)
B.E. (CSE) III-Semester (MAIN) Examinations DEC-2025/ JAN-2026

Computer Organization and Microprocessors

Time : 3 hours **Max. Marks : 60**

Note : *i) Answer **Question No. 1** (Compulsory) and answer any **four questions** from the remaining questions (2- 7). Each question caries 12 marks.*
ii) Answers must be written in same order as they occur in the Question Paper.
*iii) **Missing data, if any may suitably be assumed.***

			Marks	BT	CO
1.	a)	Show the Single precision and Double precision floating point representation with an example.	2	2	1
	b)	Compare the hard wired and micro programmed control unit.	2	4	2
	c)	What is pipeline processing? Give the classification.	2	2	3
	d)	Define addressing mode and list the addressing modes of 8085 Microprocessor.	2	2	1
	e)	List out the features of 8051 Microcontroller.	2	4	4
	f)	Draw the flag register of 8085 Microprocessor and briefly explain the same.	2	2	4
2.	a)	Design the bus system for 4 registers and explain the working of it.	6	4	2
	b)	Explain about shift micro operations and draw the 4-bit combinational circuit for the bi-directional shifter.	6	4	2
3.	a)	Describe various types of computer registers with neat diagrams.	6	2	2
	b)	Explain Interrupt Cycle with the help of a flowchart.	6	4	5
4.	a)	Classify the CPU organizations and explain the stack organization in detail.	7	2	2
	b)	Explain the operation of four-segment Instruction pipeline with suitable diagrams.	5	4	3

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5.	a)	Explain about the DMA Controller (8257) with neat diagram.	8	2
	b)	Explain any four Arithmetic instructions of 8085 Microprocessor with examples.	4	4
6.	a)	Draw the architecture of 8051 Microcontroller.	4	2
	b)	Write the program to display "WELCOME" in LCD by interfacing with 8051MC and draw the suitable hardware connections.	8	3
7.	a)	Explain about the ADC interfacing with 8051MC with diagram.	6	4
	b)	Discuss about the communication standard RS-232.	6	2

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